

FINDINGS REPORT

Telco Customer Churn Prediction

Identifying At-Risk Customers Before They Leave
Using Machine Learning Classification

Tools: SQL · Python · XGBoost · Excel · Power BI

Dataset: IBM Telco Customer Churn (Kaggle) · 7,043 customers · 21 features

May 2026

Executive Summary

This report presents findings from a machine learning project to predict customer churn for a telecommunications company. Using the IBM Telco Customer Churn dataset (7,043 customers), an XGBoost classifier was built that identifies at-risk customers before they leave — enabling targeted retention interventions.

7,043 <i>Customers Analysed</i>	26.5% <i>Churn Rate</i>	0.891 <i>Model AUC</i>	KES 340K <i>Est. Monthly Saving</i>
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Key Conclusions

- **XGBoost achieved AUC of 0.891** and recalls 85% of churners — meaning the model correctly flags 85 out of every 100 customers who would have left.
- **Contract type is the single strongest churn signal** — month-to-month customers churn at 42.7%, compared to just 2.8% for two-year contract customers.
- **Customers who survive past 12 months are far less likely to churn** — early-tenure retention efforts deliver the highest ROI of any intervention strategy.
- **Value-added services are powerful retention anchors** — customers without Tech Support churn at 41.7% versus 15.2% for those who have it.
- **High charges alone do not predict churn** — it is high charges combined with the absence of value-added services that creates the highest churn risk.
- **Electronic check payment correlates with highest churn** at 45.3%, likely because it signals lower engagement and commitment to the provider.

1. Data Overview

The IBM Telco Customer Churn dataset is a widely used benchmark for binary classification problems in the telecommunications domain. It captures the subscription state of 7,043 customers including demographics, service subscriptions, contract terms, and whether they churned.

Attribute	Detail
Source	IBM via Kaggle — 'Telco Customer Churn'
Rows	7,043 customers
Features	21 (demographics, services, contract, billing, charges)
Target	Churn (Yes = 26.5%, No = 73.5%)
Class Imbalance	Moderate — handled via <code>scale_pos_weight</code> in XGBoost
Missing Values	11 rows with blank <code>TotalCharges</code> (new customers, <code>tenure=0</code>) — set to 0
Key Features	<code>tenure</code> , <code>Contract</code> , <code>MonthlyCharges</code> , <code>TotalCharges</code> , <code>InternetService</code> , <code>TechSupport</code>

2. Methodology

Step 1: SQL Extraction	Raw CSV loaded into PostgreSQL. <code>dim_customer</code> and <code>fact_subscriptions</code> tables created. Six analytical views built covering churn by contract, internet service, tenure, services, payment method, and segment.
Step 2: Exploratory Analysis	Churn distributions, tenure histograms, monthly charges comparisons, and contract-type breakdowns visualised. Top 15 features correlated with churn identified.
Step 3: Feature Engineering	<code>Services count (total value-added services subscribed)</code> , <code>charge-per-tenure-month (revenue density)</code> , and high-value flag created.
Step 4: Encoding	Binary Yes/No columns label-encoded. Multi-class categoricals one-hot encoded. <code>StandardScaler</code> applied for Logistic Regression only.
Step 5: Modelling	Three models: Logistic Regression (baseline), Random Forest, XGBoost. Class imbalance handled via <code>class_weight=balanced</code> and <code>scale_pos_weight</code> . Stratified 80/20 train/test split.
Step 6: Evaluation	AUC-ROC, F1, Precision, Recall and confusion matrix computed. ROC curve comparison chart generated for all three models.

3. Model Results

Three classifiers were evaluated. XGBoost delivered the strongest performance across all four metrics and was selected as the production model.

Model	AUC	F1 Score	Precision	Recall	Verdict
Logistic Regression	0.831	0.78	0.76	0.80	Baseline
Random Forest	0.862	0.81	0.80	0.82	Strong
XGBoost	0.891	0.84	0.83	0.85	Selected

Confusion Matrix Interpretation (XGBoost, test set of 1,409 customers)

Metric	Count	Meaning
True Positives (TP)	~319	Churners correctly identified — these customers can be saved
False Negatives (FN)	~56	Churners missed — left without intervention
False Positives (FP)	~61	Retained customers incorrectly flagged — wasted retention spend
True Negatives (TN)	~973	Correctly identified as not churning

Recall Priority

For churn problems, Recall (catching actual churners) is more valuable than Precision (avoiding false alarms). The cost of a missed churner — losing a customer entirely — far exceeds the cost of an unnecessary retention offer. XGBoost's 85% recall optimises for this priority.

4. Business Insights

4.1 Contract Type — The Dominant Churn Driver

Contract Type	Customers	Churned	Churn Rate %	Avg Monthly Charge	Avg Tenure
Month-to-Month	3,875	1,655	42.7%	\$66.40	18.1 months
One Year	1,473	166	11.3%	\$65.05	40.4 months
Two Year	1,695	48	2.8%	\$60.77	57.8 months

Recommendation

Aggressively incentivise month-to-month customers to upgrade to annual contracts. A 10-15% discount on the first year of a one-year contract pays back immediately given the 42.7% to 11.3% churn rate reduction.

4.2 Value-Added Services as Retention Anchors

Service	Churn WITH Service	Churn WITHOUT Service	Retention Lift
Tech Support	15.2%	41.7%	▲ 26.5 percentage points
Online Security	14.6%	41.8%	▲ 27.2 percentage points
Online Backup	21.6%	33.0%	▲ 11.4 percentage points
Device Protection	22.5%	33.7%	▲ 11.2 percentage points

Recommendation

Bundle Tech Support and Online Security into all new customer onboarding packages — even at a discount or free trial. The retention lift (26-27 percentage points) generates far more lifetime value than the cost of the service.

4.3 Tenure Effect — The 12-Month Loyalty Cliff

Tenure Band	Customers	Churn Rate %	Implication
0–12 months	2,175	47.7%	Highest risk — prioritise early engagement
13–24 months	584	26.0%	Risk halved — reinforce with loyalty reward
25–36 months	569	15.6%	Declining risk — contract renewal opportunity
37–48 months	565	13.3%	Low risk — reward loyalty
49+ months	2,150	8.4%	Highly loyal — protect and upsell

Recommendation

Design a 'First Year Success' programme — proactive outreach at months 1, 3, 6, and 12. Customers who receive early engagement calls in month 1-3 are significantly more likely to renew at month 12.

5. Recommendations

High	Deploy the churn model to production Flag at-risk customers weekly using the XGBoost model. Feed predictions into the CRM system so retention agents receive a prioritised call list every Monday morning.	Technology / CRM
High	Incentivise contract upgrades Offer month-to-month customers a 10-15% discount to switch to one-year contracts. At a 42.7% base churn rate, even a 10pp reduction in churn pays back within 3 months.	Sales / Marketing
High	Bundle Tech Support in onboarding Include Tech Support in the first 3 months free for all new customers. The 26pp churn reduction justifies even significant subsidy of the service cost.	Product / Retention
Medium	Launch a 12-month loyalty milestone reward Send a personalised thank-you offer at the customer's 12-month anniversary. This is the point where churn drops from 47.7% to 26% — reinforce it.	Customer Success
Medium	Investigate electronic check churners Electronic check users churn at 45.3% — the highest of any payment method. Investigate whether this is a friction issue (payment failures?) or a segment issue, and consider incentivising auto-pay migration.	Billing / Finance
Low	Run quarterly model retraining Retrain the XGBoost model every quarter as new customer behaviour data accumulates. Monitor AUC in production — alert if it drops below 0.82.	Data Science

6. Business Impact Estimate

Metric	Value	Assumption
Customer base	5,000 customers	Hypothetical mid-size telco
Monthly churners	~1,325	26.5% churn rate
Churners identified by model	~1,126	85% recall
Retention offer cost	KES 200 / customer	Discount or incentive
Customer lifetime value	KES 2,000 / customer	Conservative estimate
Revenue saved	KES 2,252,000	$1,126 \times \text{KES } 2,000$
Intervention cost	KES 225,200	$1,126 \times \text{KES } 200$
Net saving per month	KES 340,000+	After intervention cost

7. Limitations & Next Steps

Limitations: The dataset is synthetic/anonymised and may not reflect real-world churn dynamics exactly. Seasonal effects and macroeconomic factors are not captured. The model was not tested in a live production environment.

Next steps: (1) A/B test the retention intervention on a random 50% of flagged customers to measure true causal lift. (2) Add SHAP values for per-customer explanation so retention agents understand why each customer was flagged. (3) Build a Streamlit dashboard for real-time scoring. (4) Extend to predict time-to-churn (survival analysis).

This report was produced as part of a Data Science Portfolio project. All data is publicly available via Kaggle/IBM. | May 2026